Assignment 3 : Two-Agent Tic-Tac-Toe Using Q-Learning

Problem Statement
Design and implement two Q-Learning agents — Agent 1 and Agent 2— that learn to play Tic-Tac-Toe against each other through self-play. Each agent must discover an optimal (or near-optimal) policy that maximizes its own chances of winning while avoiding invalid moves and unnecessary losses.

Description
A Tic-Tac-Toe board is a 3×3 grid of cells, each either empty, marked X, or marked O. Two independent Q-learning agents alternate turns, each learning its own Q-table purely from experience — no built-in game strategy, no min-max, no hardcoded rules beyond legality checking. Over many self-played episodes (games), both agents should converge toward strong play (ideally forcing draws against each other, since optimal Tic-Tac-Toe is a draw).

User Inputs
Board size (fixed 3×3 for classic play; kept configurable in code for extension)
Learning rate (α)
Discount factor (γ)
Exploration rate (ε)
Number of training episodes (games)
(Optional) Custom reward values

Constraints
A cell already occupied cannot be marked again (invalid move).
Game terminates when a player wins, the board fills up (draw), or the move cap is reached.
Maximum Moves = N × M = 9 (one game can have at most 9 marks placed).

Action Space
9 discrete actions — one per board cell (0–8, top-left to bottom-right).

Reward Function

Outcome	Reward
Win the game	+100
Valid move (game continues)	−1
Move into an occupied cell (invalid move)	−10
Lose the game (opponent wins)	−100
Draw	0

What You Have to Find
Optimal policies learned by both Agent X and Agent O
A sample "optimal" self-played game trace from an empty board
Total reward obtained per agent, per episode
Number of moves (steps) required to finish each game

What You Have to Show (Display)
Final Q-tables (sample of most-visited states — full table is too large to print in full)
A learned/optimal self-play game trace
Total reward per episode graph (both agents)
Steps (moves) per episode graph
Comparison of early vs. final performance (win/draw/loss rates, average game length)